Haonan Wu

Ph.D. Candidate/Research Assistant
Computer Science and Engineering at Penn State
State College, PA, USA

CONTACT INFORMATION

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EDUCATION

The Pennsylvania State University (PSU), USA — Aug. 2023 - present
Ph.D. in Computer Science and Engineering — **Research Topics**: Theoretical and Algorithmic Bioinformatics
Supervisor: Prof. Paul Medvedev

Shandong University (SDU), PRC — Sep. 2020 - Jun. 2023
Master of Science in Operational Research and Cybernetics
GPA: 92.17/100
Relevant Coursework: Graphs and Digraphs, Combinatorial Optimization, Analysis and Design of Algorithms, Modern Functional Analysis, Elements of Modern Algebra, Mathematical Analysis

Shandong University (SDU), PRC — Sep. 2016 - Jun. 2020
Bachelor of Science in Biotechnology
GPA: 90.65/100
Relevant Coursework: Biochemistry, Cell Biology, Molecular Biology

WORK EXPERIENCE

The Pennsylvania State University — State College, USA
Research Assistant in Department of Computer Science and Engineering — Jan. 2024 – present
Supervisor: Prof. Paul Medvedev

Developing bioinformatics theories and algorithms.

RESEARCH INTERESTS

I am interested in developing theories, algorithms, and practical tools for bioinformatics and addressing biological problems through these methods. My research focuses on

1. Design methods with solid math/algorithms or at least mathematical intuition.
2. Build practically useful tools with heuristics benefiting the bio/bioinfo communities.

Currently, I am working on theories and estimators of mutation and evolutionary processes based on k-mers, particularly on repetitive genomic sequences.

PEER REVIEWED PUBLICATIONS & PREPRINTS

Journals

- **Haonan Wu**, Jiyun Han, Shizhuo Zhang, Gaojia Xin, Chaozhou Mou, Juntao Liu, Spatom: a graph neural network for structure-based protein–protein interaction site prediction, *Briefings in Bioinformatics*, Volume 24, Issue 6, November 2023, bbad345, https://doi.org/10.1093/bib/bbad345
- **Haonan Wu**, and Paul Medvedev, The gift of creation: repeat-robust estimators of substitution rates, 2026 (accepted by Bioinformatics)

Conference proceedings

- **Haonan Wu**, Antonio Blanca, and Paul Medvedev. A k-mer-Based Estimator of the Substitution Rate Between Repetitive Sequences. In 25th International Conference on Algorithms for Bioinformatics (WABI 2025). Leibniz International Proceedings in Informatics (LIPIcs), Volume 344, pp. 20:1-20:20, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2025) https://doi.org/10.4230/LIPIcs.WABI.2025.20
- **Haonan Wu**, and Paul Medvedev, The gift of creation: repeat-robust estimators of substitution rates, 2026 (accepted by Intelligent Systems for Molecular Biology 2026)

Preprints

- **Haonan Wu**, Antonio Blanca, and Paul Medvedev, A k-mer-based estimator of the substitution rate between repetitive sequences, *bioRxiv* 2025.06.19.660607; doi: https://doi.org/10.1101/2025.06.19.660607
- **Haonan Wu**, and Paul Medvedev. "The gift of novelty: repeat-robust k-mer-based estimators of mutation rates." *bioRxiv* 2026.04.01.715966; doi: https://doi.org/10.64898/2026.04.01.715966

RESEARCH EXPERIENCE

Repeat-robust estimators under different settings 2026
supervised by Dr. Paul Medvedev

- Developed three k-mer-based estimators robust to repetitive genomic regions, applicable under varying levels of count information (presence-only, one-sided counts, or full counts).
- Demonstrated state-of-the-art performance within each setting, with the strongest model outperforming existing methods on highly repetitive alpha satellite sequences.
- Code available at github.com/medvedevgroup/Accurate_repeat-aware_kmer_based_estimator

Repeat-aware substitution rate estimation for repetitive sequences 2025
supervised by Dr. Paul Medvedev

- Proposed a k-mer-based estimator that relaxes the non-repetitive assumption, enabling accurate mutation rate estimation in repetitive genomes.
- Derived theoretical bounds on estimator bias and validated robustness on centromeric higher-order repeat datasets.
- Code available at github.com/medvedevgroup/Repeat-Aware_Substitution_Rate_Estimator

Spatom: structure-based protein–protein interaction site prediction 2023
supervised by Dr. Juntao Liu

- Developed a graph neural network framework modeling protein structures as weighted digraphs to capture spatial residue interactions.
- Designed a hybrid architecture combining digraph convolution and attention mechanisms, achieving state-of-the-art performance on benchmark protein complexes.
- Code available at github.com/Wu-Haonan/Spatom
- Web server at http://liulab.top/Spatom/server

ACADEMIC SERVICE

Peer Reviewer
Journals:

- Heliyon

Conferences:

- RECOMB 2025, 2026 (sub-reviewer)
- ISMB 2025, 2026 (sub-reviewer)
- ACM-BCB 2025, 2026 (primary reviewer)

MENTORSHIP & ADVISING

Undergraduate Research Mentor

- Kathryn Ellen Silverberg, Pennsylvania State University — April. 2026 - present
  Schreyer Honors Program thesis: Fast Computation of Hamming Distance Histograms for k-mers

Peer Mentorship (PhD Applications)
Advising on PhD applications, including research statements and application strategy.

- Anxuan Han — 2024
  admitted to PhD program at the University of Adelaide
- Yangyang Liu — 2025
  admitted to PhD program at the University of Southern California

TEACHING EXPERIENCE

Instructor

- Deep-learning Short Course, Shandong University — Winter 2022
  - Designed and delivered a short course on deep learning fundamentals and practical implementation with PyTorch.
  - Course materials at github.com/Wu-Haonan/Deep_learning_short_course

Undergraduate Tutor ("Dolphin Program")
Tutoring on course material, supporting students' transition to college life, and offering guidance on academic planning.

- Inorganic and Analytical Chemistry, School of Marine, SDU, China — Fall 2018
- Probability and Mathematical Statistics II, School of Marine, SDU, China — Spring 2019

TALKS & POSTER

Talks

- Title: Don't Repeat 'No Repeats':
  A k-mer-based estimator of the substitution rate between repetitive sequences
  Genome Informatics 2025, Cold Spring Harbor Laboratory, NY, USA — Nov. 5-8 2025
- Title: Don't Repeat 'No Repeats':
  A k-mer-based estimator of the substitution rate between repetitive sequences
  WABI 2025, University of Maryland, MD, USA — Aug. 20 2025
- Topic: Applications of Digraph in bioinformatics and computational biology
  Graphs and Digraphs, School of Mathematics, SDU, China — Spring 2022
- Topic: Introduction of Single Molecular Sequence and assembly algorithms for long-reads
  Sequence and Assembly Seminar, School of Mathematics and Statistics, SDU, China — Fall 2019

Posters

- WABI 2025, University of Maryland, MD, USA (Link)
- Rao Prize Conference 2025, Penn State University, PA, USA (Link)

OTHER EXPERIENCES

Explored the possible reason for abundance of oxides on the Martian surface — Jun. 2017 - Oct. 2018
Planetary Lab, Space Science School, SDU with Dr. Zhongchen Wu

SKILLS

- **Programming:** Python, MATLAB, C++, experience programming in Linux server
- **Package:** Pytorch, scikit-learn
- **Scripting Language:** LaTeX
- **Language:** Chinese (native speaker), English (working language)

REFERENCES

Prof. Paul Medvedev
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